PES University — Dept. of CSE

SOFTWARE ENGINEERING LAB 3

Component Modelling & Architectural Pattern Selection

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Scenario — Self-Service Coffee Kiosk System

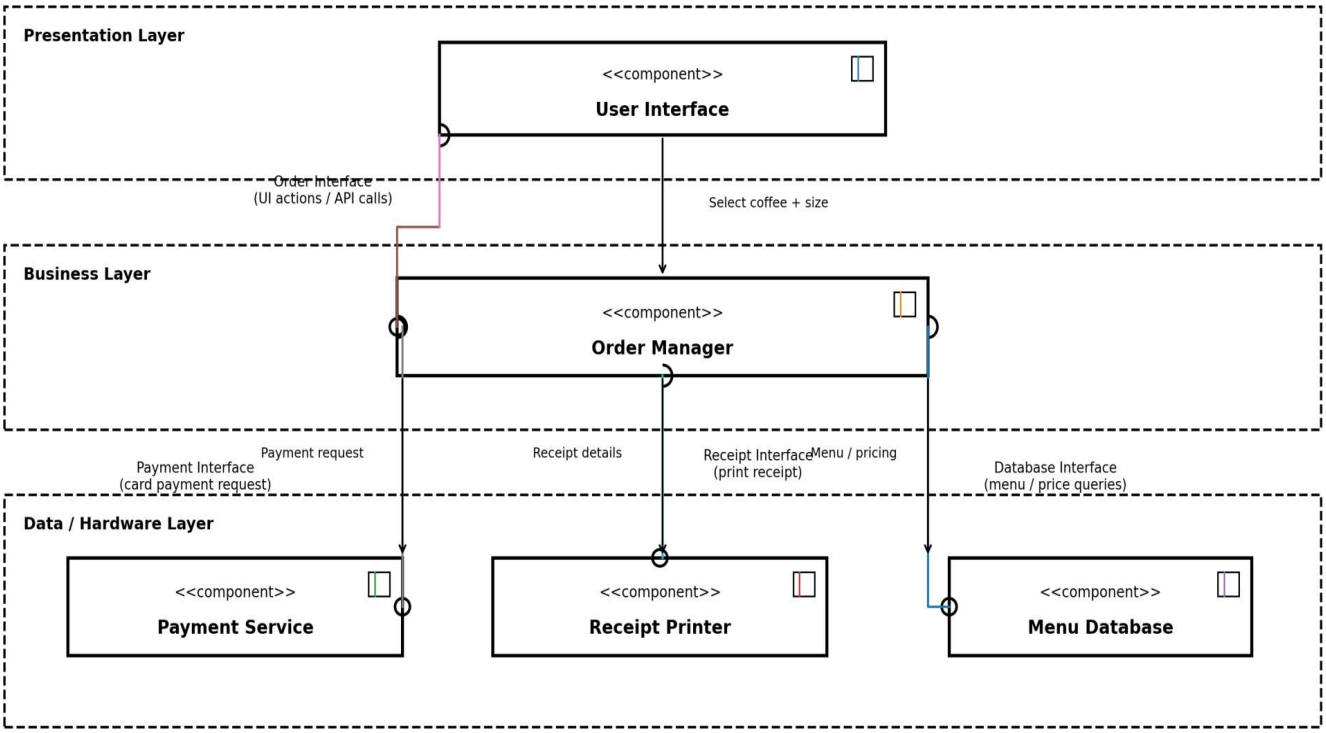
A self-service cafe kiosk that lets customers select a coffee type and size, pay using a credit card, print a receipt, and access menu and pricing information.

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1. UML Component Diagram

Self-Service Coffee Kiosk System — Layered Architecture



2. Written Justification

Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Comparison of Architectural Styles

Layered architecture provides clear separation between the presentation, business, and data/hardware responsibilities. Micro-services would provide independent deployment and scaling, but would add operational and network complexity that is unnecessary for a single kiosk. Client-server is simple, but its centralized design can create a bottleneck or single point of failure.

Two Reasons for the Choice

1. Clear separation of responsibilities: the User Interface handles touch-screen interaction, the Order Manager handles ordering logic, and the Payment Service, Receipt Printer, and Menu Database handle specialized functions. This makes the kiosk easier to understand and maintain.

2. Suitable for the kiosk's requirements: the system has a small and fixed set of coffee types, sizes, credit-card payment, receipt printing, and menu/pricing storage. A layered design provides these functions without the additional deployment complexity of micro-services.

Security Advantage

The Payment Service is separated from the user-interface and order-management logic. Payment requests can pass through a controlled interface, helping isolate sensitive card-processing responsibilities from the rest of the kiosk.

Performance Benefit

The kiosk can use direct component communication and local menu/pricing data access, avoiding unnecessary distributed-service calls. This supports quick touch-screen responses and fast order processing in a busy cafe.

Components and Interfaces

The diagram contains five components: **User Interface, Order Manager, Payment Service, Receipt Printer,** and **Menu Database**. It shows four main interfaces: **Order Interface, Payment Interface, Receipt Interface,** and **Database Interface**, using provided-interface (ball) and required-interface (socket) notation.